



Sana'a University
faculty of Computer and information Technology

layered mitigation for Neighbor Discovery Protocol security in Dual-stack cloud environment

Done by :Maram M. Al:shahari

supervisor:
D/ Ahmed Alshalabi

October 13, 2026
cybersecurity

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Chapter 1 : Introduction

1 Introduction

As technology continually evolves cloud computing has become the backbone of modern digital infrastructure, with major suppliers in both IPv4 and IPv6.The exhaustion of IPv4 addresses has urged cloud operators to deploy IPv6 alongside IPv4 in a dual-stack architecture, where every virtual machine runs both protocol stacks simultaneously. While dual-stack enables gradual migration,it introduces profound security challenges that are frequently underestimated. One of the most critical challenges is the Neighbor Discovery Protocol (NDP), the IPv6 equivalent of ARP, router discovery, and redirect functions combined. NDP operates at the link layer and is essential for IPv6 communication, but it was designed without any mandatory authentication or encryption mechanisms. The original specification assumed that all nodes on the same link are trusted, a reasonable assumption in isolated private networks, however cloud environments where thousands of tenants share the same physical infrastructure. This research addresses the problem of private dual-stack cloud environment, an attacker who gains access to any machine on the same link can forge Router Advertisement messages to become a man-in-the-middle(MITM), flood the network with Neighbor Solicitation or Neighbor Advertisement messages to cause denial of service, exploit Duplicate Address Detection to prevent legitimate devices from obtaining IPv6 addresses, and replay old NDP messages to poison neighbor caches. These attacks can lead to data interception, service disruption, and tenant isolation, directly undermining cloud security guarantees. Nevertheless, no single mitigation has emerged as a clear, practical solution for cloud-scale dual stack environments .Secure Neighbor Discovery (SeND) suffers from high computational overhead ,cryptographic protocol (NDPsec) lacks widespread adoption, Voucher based-address(VBA) alone provides proactive verification, and anomaly based intrusion detection using entropy and neural networks is not enough alone as a reactive countermeasure. Thus, this research proposes a novel countermeasure by designing a layered cryptographic verification system combining VBA with Backpropagation Neural Network (BPNN) anomaly detection to maximize detection while minimizing computational overhead, and to determine the optimal balance between proactive verification and reactive detection, providing cloud architects and security engineers with actionable guidance to secure NDP in dual stack cloud deployments.

1.1 Research Problem

The coexistence of IPv4 and IPv6 in dual-stack clouds creates additional attack surfaces such as spoofing(forging NS/NA messages) and flooding (excessive RA messages) which lead to data interception, tenant isolation breaches, service disruption. There are several mitigation techniques, including Secure Neighbor Discovery (SeND), cryptographic protocols (NDPsec), and Voucher based-address(VBA) to anomaly based intrusion detection using entropy and neural networks. However, (SeND) suffers from high computational overhead ,(NDPsec) lacks widespread adoption, (VBA) alone provides proactive verification, and anomaly based intrusion detection using entropy and neural networks is not enough alone as a reactive countermeasure. Thus, no single mitigation has emerged as a clear, practical solution for private dual stack cloud environments . therefore, research proposes a novel countermeasure by designing a layered cryptographic verification system combining Voucher based-address(VBA) with Backpropagation Neural Network (BPNN) anomaly detection to maximize detection while minimizing computational overhead.

1.2 Purpose Statement

The purpose of this experimental quantitative research is to design and empirically evaluate a layered cryptographic verification system VBA combining with BPNN anomaly detection for A private dual-stack cloud environment. To determine the optimal balance between proactive cryptographic verification(VBA) and reactive ML- based detection(BPNN) to minimize the NDP attack risk.

1.3 Research Questions

1.3.1 Main RQ

RQ1: How can a layered cryptographic verification system ,combining anomaly detection with a lightweight machine learning model ,be optimized to maximize detection while minimizing computational overhead for mitigating NDP spoofing and flooding attacks in resource -constrained dual-stack cloud environment ?

1.3.2 Sub RQs

RQ2: How can the network and Voucher Based Addressing (VBA) be integrated with centralized backpropagation neural network (BPNN) detection engine to enable a dynamic and real-time layered defense against NDP spoofing and flooding attacks in dual-stack cloud environment?
RQ3: What is the optimal balance between proactive cryptographic verification(VBA) and reactive ML- based detection backpropagation neural network (BPNN) to minimize NDP attacks risk while managing computational overhead in the dual-stack cloud environment ?

1.3.3 Research Objectives

1.3.4 Main objective

To design and empirically evaluate a layered cryptographic verification system combining VBA with BPNN anomaly detection for A private dual-stack cloud environment. To test the optimal balance between proactive cryptographic verification Voucher Based Addressing (VBA) and reactive ML- based detection backpropagation neural network (BPNN) to minimize the NDP attack risk.

1.3.5 sub objectives

1- To optimize the system for detection to maximize detection while minimizing computational overhead.
2- To determine the optimal balance between proactive cryptographic verificationVoucher Based Addressing (VBA) and reactive ML- based detection backpropagation neural network (BPNN) to minimize the NDP attack risk.
3- To evaluate the Protection rate, Detection Rate, and CPU overhead.

1.4 Significance of the study

This quantitative study is significant because it focuses countermeasures of IPv6 Neighbor Discovery Protocol (NDP) attacks within dual-stack environments. It directly serves network architects and security engineers who are responsible for designing modern countermeasure to mitigate network attacks at several levels. At the theoretical level, it develops a layered defense frame work layering proactive cryptographic verification Voucher Based Addressing (VBA) with reactive ML-based detection backpropagation neural network (BPNN). Providing quantitative evaluating metrics(Protection rate, Detection Rate, and CPU overhead). Establishing a theoretical understanding the optimal balance between proactive and reactive model. At the practical level, the research provide a practical benefits to cloud architects and security engineers who need actionable guidance to secure NDP in dual stack cloud deployments. Cloud service providers who wants to reduce risks of data interception and breaches. At the methodological level, the study addresses methodological approaches by using network simulation (GNS3) with 15 runs to evaluate NDP defense system in dual-stack cloud environments. The research enables future studies by opening new research questions about the optimal layering of cryptographic and ML-based defense

1.5 Scope / delimitation

The scope of this quantitative study is focuses on dual-stack cloud environments across small-to-medium network architectures. It focuses on IPv6 Neighbor Discovery Protocol (NDP) known vulnerabilities, attack vectors. This study does not include large network infrastructures cloud or extensive corporate networks environments. This study conduct using network simulation (GNS3) with 15 runs, it does not involve real production cloud infrastructure or physical deployment. This delimitation is intended to keep the study focused on current NDP attacks(2024-2026),it does not beyond 2026.

1.6 Limitation

This quantitative study has several limitations. First, uses GNS3 simulation with 60 runs instead of real production cloud traffic which may not replicate real cloud performance. Second, the countermeasures that will be combined were not applied in cloud environment before,thus the percentage of success in cloud environment is not accurate. Last, GNS3 has limited IPv6 feature support, potentially affecting simulation accuracy.

1.7 Definition of terms

To ensure clarity, technical accuracy in this study as follows:
1- Ipv6(internet protocol version 6): is the most recent version of Internet Protocol (IP). It's designed to supply IP addressing and additional security to support the predicted growth of connected devices in IoT, manufacturing, and emerging areas like autonomous driving.
2- Daul-stack : Dual Stack refers to the capability of supporting both Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6) simultaneously on a network or system.
3- Neighbor Discovery Protocol (NDP): is a protocol in the IPv6 suite designed to facilitate communication between devices on the same network. NDP replaces the Address Resolution Protocol (ARP) used in IPv4, offering more functionality and efficiency. NDP uses Internet Control Message Protocol for IPv6 (ICMPv6) messages to perform its tasks, including address resolution, neighbor reachability, and address autoconfiguration.

- NDP Messages :

• Router Solicitation (RS): Sent by a device to request information about available routers on the network.
• Router Advertisement (RA): Sent by routers to provide information about themselves, prefixes, and configuration parameters.
• Neighbor Solicitation (NS): NS messages are used to request a device's MAC address or verify its reachability.
• Neighbor Advertisement (NA): Devices respond to NS messages with NA messages, providing their MAC addresses or confirming reachability.
• Redirect: Sent by routers to inform devices about a better route for a specific destination. This message helps optimize routing and reduce latency.

4- Attack vector: is the specific path or method a threat actor uses to breach your network or system.
5- Denial-of-Service(DOS): deliberately exhausts the availability of a system, service, or network resource by flooding it with requests or exploiting operational bottlenecks.
6- Man-in-the -Middle(MITM) : is a general term for when a perpetrator positions himself in a conversation between a user and an application—either to eavesdrop or to impersonate one of the parties, making it appear as if a normal exchange of information is underway.
7- Spoof: a type of cybercriminal activity where someone or something forges the sender's information and pretends to be a legitimate source, business, colleague, or other trusted contact for the purpose of gaining access to personal information, acquiring money, spreading malware, or stealing data.

8- Countermeasure: Actions, devices, procedures, techniques, or other measures that reduce the vulnerability of an information.

• architectural countermeasures :network-level security configurations and strict port binding controls.

• cryptographic protocols : security techniques leveraging cryptographic primitives, digital signatures, and public key infrastructures

1.8 Brief Methodology Overview

Developing a dual-stack cloud environment introduces vulnerabilities to many attacks ,one of the attacks is Neighbor Discovery Protocol (NDP) , while several methodologies have been addressed in chapter 2 to mitigate NDP attacks , only a minority of them succeeded to protect from the most of the attacks. This quantitative experimental proposal aims to answer the research questions in section(1.3) that focuses on suggesting new model that mitigate NDP attacks properly by addressing a layered methodology that combines reactive and proactive methodologies minimize the possibility of attacks to occur by using simulation and dataset to do experiments analysis.

1.9 Organization of the Thesis

Chapter 1 – Introduction
Chapter 2 – Literature Review
Chapter 3 – Methodology / Research Design

This thesis is organized into five chapters. Chapter 1 introduces the study by presenting the background, problem statement, purpose, research questions, objectives, significance, scope, limitations, and definition of terms. Chapter 2 reviews the relevant literature on [main topics], highlighting key theories, models, and empirical studies related to [your constructs]. Chapter 3 describes the research methodology, including the research design, population and sample, instruments, data collection procedures, and data analysis methods. Chapter 4 presents the results of the data analysis and, where appropriate, answers the research questions. Finally, Chapter 5 discusses the findings in light of the literature, draws conclusions, outlines the implications of the study, and offers recommendations for practice and future research .

Chapter 2 : Literature Review

2 Introduction

Despite IPv6's security improvements over IPv4,practicaly in dual-stack cloud environment the gap of security is not covered . one of the common gap is NDP insecurity while the Neighbor Discovery Protocol (NDP) is essential for IPv6 link-local communication ,it was designed without authentication or integrity checks, which make it vulnerable to many attack vectors. The recent literature proposed countermeasures strategies to mitigate NDP attack, but they failed to balance between performance, delay, complexity, and to protect against as many attacks as possible. To address this limitation, this study aim to propose proactive and reactive method that enhance two security mechanism to merge it in daul-stack cloud environment The remainder of this chapter is organized as follows: section 2.2 explores the theoretical foundation of NDP vulnerabilities and attacks. Section 2.3 provides a critical review of existing proactive and reactive countermeasures, finally section 2.4 synthesizes the identifies the research gap to justify the proposed solution.

2.1 Theoretical background

2.1.1 Foundational Vulnerabilities of NDP

The Neighbor Discovery Protocol (NDP) is a core component of IPv6, yet its original design (RFC 4861) lacks fundamental security mechanisms. Specifically, NDP does not require authentication, does not verify message integrity, and is stateless, which makes it inherently vulnerable to a range of attacks [6].

vulnerability	Description	Source
No authentication	Any node on the link can send NDP messages (RA, NS, NA, RS) without proving its identity	[1],[6]
No integrity verification	Messages can be modified in transit without detection	[1]
Stateless operation	No session tracking makes replay attacks trivial	[4]
DAD trust assumption	During Duplicate Address Detection, the first NA response is accepted as true, allowing address claiming	[1]
RA/RS origin verification is missing.	Any node can impersonate a router.	[6]
SLAAC privacy weakness	EUI 64 address generation embeds MAC addresses, enabling device tracking.	[5]

Table 1. Inherent vulnerabilities of NDP

These flaws are particularly dangerous in cloud environments where multiple tenants share the same physical link [2].

2.1.2 NDP Attacks

Based on [1][2][6][7] the most frequent NDP attacks are:

Attack Type	Description	Impact	tool	source
RA Flooding	Thousands of rogue Router Advertisements	CPU exhaustion, DoS	flood$_r outer6(fromTHCIPv6toolkit)$	[2],[6]
NS Flooding	A massive number of Neighbor Solicitation messages.	Neighbor cache overflow, system freeze	flood$_s olicitate6(fromTHCIPv6toolkit)$	[2],[6]
NA Spoofing	Rogue Neighbor Advertisement with attacker's MAC address.	Man in the middle, traffic redirection	Scapy (crafted NA messages)	[1]
DoS on DAD	Attacker responds to a DAD NS claiming the tentative address is already in use.	The host cannot obtain an IPv6 address	flood$_a dvertise6(THCIPv6) and Scapy$	[1]
Redirect Spoofing	Rogue Redirect message changes the routing path.	Routing manipulation, MITM	No specific tool mentioned	[6]
Replay Attack	Old NDP messages are resent.	Cache poisoning, DoS, MITM	No specific tool mentioned	[4]

Table 2. Common NDP attacks and their impact

2.2 critical review of existing proactive and reactive countermeasures

This section categorizes the existing countermeasures that are identified in recent studies into two primary groups and critically evaluates them, and assesses where the studies fail to strike an optimal balance between security performance, complexity, latency, and attack prevention or detection.

2.2.1 Proactive countermeasures

2.2.1.1 IPsec (Original IPv6 Specification – 1998)

IPsec can only be used with a manual configuration of security associations. This is due to bootstrapping problems in using IKE, which is the protocol used to establish IPsec security associations. Additionally, the number of manually configured security associations needed for protecting NDP can be very large[6] making this approach impractical for most purposes.

2.2.1.2 Secure Neighbor Discovery (SEND) and Cryptographically Generated Addresses (CGA) – 2005

SEND was developed as a protocol extension to add security features to NDP by using Cryptographically Generated Addresses (CGA) to encrypt NDP messages. CGAs are generated using a public-key infrastructure (PKI), which allows nodes to verify the authenticity of each other's CGAs. This prevents attackers from spoofing NDP messages or launching other attacks against the NDP. However, the SEND is not widely deployed due to several reasons:

1- Compatibility: SEND is not backward compatible with existing IPv6 devices and networks. Therefore, deploying it requires a complete overhaul of the network infrastructure.

2- Complexity: The implementation of SEND is complicated and requires additional resources and expertise. This can increase the cost of deployment and maintenance because it requires additional resources, such as public key infrastructure (PKI), which adds to the cost of implementation.
3- Lack of awareness: Many network administrators and users are not aware of the vulnerabilities present in the NDP protocol. They also do not know about the benefits of using the SEND protocol[6].
SEND is vulnerable to flooding attacks because processing RSA signatures consumes significant CPU resources, allowing an attacker to overwhelm a host with signed messages. Additionally, SEND and CGAs have not been widely deployed due to high complexity, intellectual property claims, and licensing terms[4]. Additionally, SeND aims to prevent NS and NA spoofing only with high latency and complexity during the RSA cryptography

2.2.1.3 Match Prevention – Lightweight Hashing for DAD and AR, and Solving the EUI 64 Privacy Problem[5]

Match-Prevention is designed to secure AR and DAD against DoS attacks in an IPv6 link-local network by preventing the disclosure of target IP addresses during the AR and DAD processes, securing NS and NA messages independently , and evaluating the performance. proposed technique comprises three main stages:-
satge 1: To achieve this goal, the hashing mechanism is applied to hide the address for security purposes. Firstly, all the hosts should generate their IP addresses by using the privacy extension method instead of the EUI-64 bit method. The privacy extension method randomly generates the Interface ID (IID) of an IPv6 address, thus helping to prevent an attacker from predicting the IP address and maintaining the privacy of the host. moreover the method of encryption that it uses SHAKE128 given its 128-bit output. In this way, the proposed technique consumes less processing time for extensive calculations during the generation verification of NS and NA messages through the AR/DAD processes in an IPv6 link-local network. However, using the sole IID (64 bits) as an input 'plaintext' for SHAKE128 is vulnerable to brute-force attacks.
Stage 2:Match-Prevention introduces NS-match and NA-match messages without changing the original structure of NDP but uses the NDP experimental option The experimental NDP option, match-option, comprises three main fields, namely Nonce, RIN and IPhash, and aims to distinguish valid NS/NA messages from bogus ones. The match-option is appended into each NS and NA message to transform them into NS-match and NA-match messages, respectively, during AR and DAD. Each message without the match-option should be discarded .it uses a rule based mechanism that allows hosts to verify NS and NA messages independently, without third party intervention.
However,it is lightweight but, the technique secured only DAD and AR attacks , leaving RA, RS, and Redirect messages unprotected ,this incomplete coverage left a clear gap for more comprehensive solution.

2.2.1.4 NDPsec [4]

Instead of using EUI-64 or Privacy Extension.The NDPsec algorithm derives the IID for a particular host using the public key of a digital signature. Each IPv6 host that joins the network through the proposed NDPsec mechanism is required to generate public and private key pair using a digital signature algorithm. The Ed25519 digital signature algorithm is selected for this research because it generates the public and private key pair fast and has a smaller public key size than other digital signature algorithms. It is designed to be faster than existing digital signature schemes while maintaining the same level of security. The NDPsec mechanism takes advantage of this by treating LF as an IID and uses it as an IPv6 address for the host intending to join the network. The RF is subsequently used in conjunction with the IID (i.e., LF) to reconstruct the public key on the receiving hosts, which is used to validate the integrity of the message. The standard NDP messages have the option field, which can be used to append other useful information to convey the digital signature and the LF. Secure-option consists of four main fields, namely, RDM, RDM-Info, LF, and DS. The primary objective of proposing this option is to differentiate legitimate NDP messages from invalid ones. Then sender of the IPv6

host signs the NDP message. The NDPsec messages must be digitally signed with the private key generated in step one to prevent them from being altered in transit or spoofed by threat actors in order to launch cyber-attacks. Next, verify the NDP message by validating the digital signature that was included in the message by receiving host must first extract the public key from the message. NDPsec achieves this by combining IID with the RF value contained within the message. In addition, NDPsec makes use of the RDM and RDM-Info fields of the Secure-option to prevent replay attacks. Nonce and timestamp are the two prevalent ways of preventing replay attacks. However, the technique still interduces processing latancy and bandwith overhead because of the message size , and solved only attack DAD, RA, Redirect attacks.

2.2.1.5 Voucher-Based Addressing (VBA) – Solving Neighbor Spoofing and Privacy Threats with a Low Complexity, Privacy Preserving Alternative [7]

VBAs will always appear random as a consequence of utilizing the outputs of deterministic hash functions. Finally, using key derivation techniques as the deterministic hashing functions introduces an adjustable work factor that modifies VBA generation requirements and mitigates brute-force attacks. this work elevates simple hashing techniques to the use of key derivation functions (KDFs), KDFs accept input iteration values specifying how many times the driving pseudo-random function (or underlying hash function) must be iterated before producing a final result. In VBA generation, various inputs that specifically identify the target node, as well as the chosen iteration count, are computed by a known KDF into a hash value. It done by Router X which is a Voucher Bearer authorized by local administrative policy to delegate Link Voucher information to its neighbors such as(secret seed, hash algorithm to use, how many times to hash), then host receive a voucher and pick an iteration count, take its MAC address +voucher seed +VBA and run hash function ,the result is 48 bit number the address suffix. the receiver extract the encoded iteration count use the same voucher secret and provided a MAC to recompute the hash if it match ,it is valid. In addition, VBA is less complex while there is no PKI or digital signature is required, moderate performance and high privacy ,however, it does not stop DOS, does not work for anycast, it breaks if the prefix is longer than/64, finally, it needs a trusted router (use RA guard to prevent fake voucher.

2.2.2 reactive countermeasures

2.2.2.1 Entropy Based Flow Detection – Solving the Problem of Undetected RA and NS Flooding Attacks [2]

It detect RA and NA flooding attacks by first, capture and filter only ICMPV6 messages types 134,135, second, build a flow which is a packets that has the same features such as(source IP address, destination IP address,ICMPV6 type, network prefix),third, measure the randomness which called entropy of source IP address and network prefixes. forth,the value compare to static threshold set to 1.,if the entropy exceeds the threshold for three second ,an flooding attack is detected. fifth, if the rule trigger ,the system log an alert. It is 100% no false negative,and the precision is 55%. However, this method depend in static threshold which call some normal traffic an attack, it detects only ICMPv6 flood attack.

2.2.2.2 Neural Network Based Anomaly Detection – Solving the Problem of Low Detection Accuracy and High False Positives in Rule Based Systems, but Limited to RA DDoS Attacks [8]

Neural Network Based Anomaly Detection is a reactive method that uses Backpropagation Neural Network (BPNN) to let the network learn by trial and errors by feeder with thousands of RA packets that are already known as normal, and others known as attack. Then deploy the trained network and let it filter traffic and extract 5 features for each packet(time,source IP, destination IP address, protocol type, and packet length. The dataset is balanced using STORM oversampling technique to avoid bias. The system achieved 99.95% detection accuracy ,99.92% precision,99,98 recall ,and false positive rate of 0.04% for RA flooding ,DDOS attacks, but

this method ignore the rest of attacks ,and require labeled training dataset, and the attack generates a dense burst of identical size packets to a single destination. A slow, low rate attack might evade detection.

2.3 Research synthesis and identification of the research gap

This paper discusses five cryptography strategies:
First, IPsec is considered the basic strategy to mitigate Neighbor Discovery Protocol(NDP) threats, but it needs manual configuration, which is a drawback since it is not a default configuration.
Second, the use of a combined Cryptographically Generated Address(CGA) and Public key to improve the encryption security. However, it is vulnerable to flooding attacks because processing RSA signatures consumes CPU resources. This allows an attacker to overwhelm a host with signed messages. this method also faces high complexity, intellectual property claims, and licensing terms challenges as well.
Third, match-prevention strategies secure Duplicate Address Detection (DAD) and prevent the exposure of EUI-64 interface addresses by prompting the host to use the privacy extension method. This technique is considered lighter than Secure Neighbor Discovery (seND), but it only secures DAD and address resolution(AR). It does not protect Router Advertisement (RA), Router Solicitation (RS), or Redirect messages, which is considered a huge limitation and can't fulfill to secure NDP.
Fourth, NDPsec is viewed as the most effective strategy to overcome the gaps that are presented in IPsec, seND, and match-prevention by using a digital signature for authentication, and reduces the computational cost compared to RSA, however, its limitations are the small message size and the need for a router to distribute the keys. One of the suggestions is to use KDC (key distribution system) such as PKI, PRE-SHARED KEY, or Kerberos.
Fifth, the simplest, which is both flexible and secure, is (VBA) that mitigates neighbor spoofing attacks and privacy threats caused by EUI-64 address generation . Additionally, it helps limit the redirection attack, while it is robust against brute force attack, it isn't a complete replacement for NDP authentication in the cloud.
On the other hand, there are three detection-based strategies to consider :
First, there is flow-based entropy detection that monitors (RA) and (NS) flooding in real time .this approach involves filtering ICMPv6 messages and setting an entropy compared to a static threshold set to 1 as well . while This strategy is accurate and reduces the false positives, it is reactive in nature ,if an attack occurs concurrently ,it can be costly for the organization a lot if an attack happens beside it, as it relies on static threshold.
Second, a neural network-based anomaly detection system that overcomes the dependency on static thresholds and utilizes a backpropagation neural network(BPNN) to classify traffic as either normal or anomalous in (RA) and (DDoS), but totally ignores redirect spoofing, DoS in DAD, and needs a normalized data set to function effectively.
it's crucial to note that relying solely on a reactive strategy is one of the worst decisions an administrator can make; the key is to adopt a proactive strategy supported by a reactive system. One potential solution is to combine a backpropagation neural network strategy and NDPsec to maximize the protection, and both are considered lightweight and highly efficient, maximizing protection.
Finally, a significant gap remains because none of the cryptographic strategies have been tested against the vast amounts of traffic typically found in dual-stack cloud environments, or think about a solution if quantum computing becomes available because (RSA,ECC) is vulnerable, NDPsec can be adapted if they enhance the hash-based signature, such as (SPHINCS+), VBA likely can't relies on traditional public-key crypto. Furthermore, public cloud computing will either require a hypervisor for NDP authentication, or an overlay network that encapsulates and authenticates all neighbor traffic called (VXLAN), The aim propose solution is to combine two strategies, first, enhancement of Voucher-based addressing (VBA). Second, detective strategy to be in pc the has mirror port to switch , to achive comprehensive attack mitigation while preserving virtualization performance, reducing communication latency, and minimizing network complexity .

2.4 Literature Matrix

REFRENCE	CONTEXT	METHOD	WHAT THEY DID	FINDING	LIMITATION	RELEVANCE
AHMED K. AL-ANI , MOHAMMED ANBAR , AYMAN AL-ANI, AND DYALA R. IBRAHIM. (2020)	National Advanced IPv6 Centre, Universiti Sains Malaysia, Gelugor 11800, Malaysia	Article,+ experimental	Match-Prevention is proposed to prevent DoS attacks during AR and DAD in a IPv6 link-local network by hiding the target IP address using the SHA-3 hash-function algorithm. Security and comparative analyses are carried out on existing techniques	1- consumes less processing time and bandwidth consumption and is more efficient compared with SeND and Trust-ND. 2- compare its results of two scenarios. 3- resists various types of attacks, such as collision and brute-force attacks. 4- efficiently prevent DoS attacks during AR and DAD.	1- implemented on a small-area IPv6 network	1- Does not apply it in large network environment or cloud. 2- the technique secured only DAD and AR attacks , leaving RA, RS, and Redirect messages unprotected
1 ABDULLAH AHMED BAHASHWAN , MOHAMMED ANBAR 2 IZNAN HUSAINY HASBULLAH 1, ZIYAD R. ALASHHAB 1, AND ALI BIN-SALEM	1 National Advanced IPv6 Centre (NAv6), Universiti Sains Malaysia, Penang 11800, Malaysia 2 School of Computer Science, Neijiang Normal University, Neijiang 641100, China	Article,+ experimental	Entropy based-flow method highly capable of detecting abnormal NDP traffic caused by RA and NS DoS flooding attacks	1-entropy algorithm to calculate the randomness of source IP address and network prefix, which are considered the clues to detect abnormal NDP traffic behavior. 2- 98.1%, 55%, 100%, and 70.96% for the average accuracy,precision, recall, and F1-Score, respectively, for the detection of the abnormal NDP traffic behavior caused by the RADoS flooding attack 3- obtained 99%, 91.3% 100%, and 95.45% for the average accuracy, precision, recall, and F1-Score, respectively.	1- needed to use various techniques such as dynamic/adaptive threshold algorithm. 2- small network experiment	1- method depend in static threshold which call some normal traffic an attack, it detects only ICMPv6 flood attack. 2- Does not apply it in large network environment or cloud

Table 1: Table3: Literature Matrix

REFRENCE	CONTEXT	METHOD	WHAT THEY DID	FINDING	LIMITATION	RELEVANCE
AYMAN AL-ANI , AHMED K. AL-ANI , SHAMS A. LAGHARI , SELVAKUMAR MANICKAM , KHIN WEE LAI (2022)	1Faculty of Computing and Informatics, Universiti Malaysia Sabah, Kota Kinabalu, Malaysia 2School of Computing and Data Science, Xiamen University Malaysia, Sepang, Selangor, Malaysia 3National Advanced IPv6 Centre (NAv6), Universiti Sains Malaysia (USM), Gelugor, Penang 11800,Faculty of Engineering, University of Malaya, Kuala Lumpur 50603, Malaysia	Article+ expermental	NDPsec new mechanism called NDPsec that enables authentication for NDP communications in an IPv6 link-local network.	The experiments show clear superiority in NDPsec in most of the experiments as compared to SeND Match-Prevention, and Trust-ND. Therefore, NDPsec is considered a better alternative than SeND Match-Prevention, and Trust-ND	1- Do the experment in small local network 2- need proposing a mechanism to distribute the router's public key in the link-local network.	1- the technique still interduces processing latency and bandwidth overhead because of the message size , and solved only attack DAD, RA, Redirect attacks. 2- Does not apply it in large network environment or cloud

Table 2: Table3: Literature Matrix

REFRENCE	CONTEXT	METHOD	WHAT THEY DID	FINDING	LIMITATION	RELEVANCE
Naveen Kodakandla (2023).	Independent Researcher, Aldie, Virginia, USA.	Review article	differences between IPv4 and IPv6 from the aspects of their performance, cost, and security	1-IPv4 has been key in the building of internet connectivity, the lack of scalability . 2-security vulnerabilities and reliance on clumsy workarounds like NAT make for an urgent need for a robust replacement 3- IPv6 is key role in the next generation of cloud computing lies in its ability to accommodate such emerging technologies as IoT, 5G and AI driven networks		
Puhl, Z.T.; Guo, J. Securing (2024)	College of Engineering	Computer Science, University of Michigan-Dearborn, Dearborn, MI 48128, USA	Article + experimental	VBA is to define an alternative to SEND,CGAs, and other monolithic approaches to solving the neighbor spoofing problem	1-Achieve privacy,flexibility, and simplicity 2-defined a decentralized and empirical approach that mostly avoids the aforementioned tar pits and creates a new perspective3-secures IPv6 Neighbor Discovery address resolution (NDAR) by cryptographically binding a link layer address (LLID) to an IPv6 address without modifying the NDP protocol itself	1-VBA is not compatible with network whose subnet prefix exceeds 64 bit.2- anycast addresses can't be verified.1-Solve spoofing attack only 2-Does not apply it in large network environment or cloud

Table 3: Table3: Literature Matrix

REFRENCE	CONTEXT	METHOD	WHAT THEY DID	FINDING	LIMITATION	RELEVANCE
Saad, R. M. A., Anbar, M., Manickam, S., Shaheen, S. I.,Hasbullah, I. H. (2024)	The Author(s), under exclusive licence to Springer-Verlag GmbH, DE 2024	REGULAR CONTRIBUTION	detection system called anomaly-based detection system using neural network algorithm	1-The experimental results revealed that the system attained an average accuracy of 99.95of 99.92%, recall of 99.98%, and F1-Score of 0.04 in detecting abnormal NDP traffic resulting from DDoS anomaly attacks.	1-capability to exclusivrly detect NDP anomalous resulted from RA-based attacks at the network layer through the monitoring of anomalous network traffic behavior using BPNN	1-ignore duplicate address detection attacks and Spoofed ICMPv6 Neighbor Advertisement 2-Doesn't apply it large network environment or cloud
Firas Najjar *, Qusay Bsoul and Hasan Al-Refai (2023)	Information Security and Cybersecurity, Philadelphia University, Amman 19392, Jordan;	article	testbed is used to analyze the impact of NDP attacks on networks and operating systems.	1- network that allows for digital signatures during the IPv6 neighbor discovery process can effectively prevent attacks like the falsification of RA and NS messages. Nonetheless, implementing this approach may not be practical for networks with a large number of users or that are publicly accessible 2-crucial to monitor network activity to identify any deviations from the normal behavior of the NDP protocol, such as blocking messages that are outside of the protocol's expected behavior.To achieve this, network tools can be utilized to detect and prevent potential attacks, such as firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS).	1-tools must be configured, probably because misconfiguration could impact the normal behavior of the protocol.	1- Does not discuss detective system

Table 4: Table3: Literature Matrix

REFRENCE	CONTEXT	METHOD	WHAT THEY DID	FINDING	LIMITATION	RELEVANCE
Kwun-Hung Li and Kin-Yeung Wong (2021)	School of Science and Technology, The Open University of Hong Kong, Hong Kong, China;	REGULAR CONTRIBUTION	detection system called anomaly-based detection system using neural network algorithm	1-The experimental results revealed that the system attained an average accuracy of 99.95of 99.92%, recall of 99.98%, and F1-Score of 0.04 in detecting abnormal NDP traffic resulting from DDoS anomaly attacks.	1-capability to exclusivrly detect NDP anomalous resulted from RA-based attacks at the network layer through the monitoring of anomalous network traffic behavior using BPNN	1-ignore duplicate address detection attacks and Spoofed ICMPv6 Neighbor Advertisement 2-Doesn't apply it large network environment or cloud
Hamdou, M., El Haloui, M., El Ksimi, A., Ettaki, B. (2025))	IResearch Laboratory of Informatics, Data Sciences and Artificial Intelligence, School of Information Sciences, Rabat, Morocco	article	testbed is used to analyze the impact of NDP attacks on networks and operating systems.	1-network that allows for digital signatures during the IPv6 neighbor discovery process can effectively prevent attacks like the falsification of RA and NS messages.Nonetheless, implementing this approach may not be practical for networks with a large number of users or that are publicly accessible .	1-tools must be configured, probably because misconfiguration could impact the normal behavior of the protocol.	1- Does not discuss detective system

Table 5: Table3: Literature Matrix

Chapter 3: Research Methodology

3 Introduction

Developing a dual-stack cloud environment introduces vulnerabilities to many attacks ,one of the attacks is Neighbor Discovery Protocol (NDP) , while several methodologies have been addressed in chapter 2 to mitigate NDP attacks , only a minority of them succeeded to protect from the most of the attacks. This quantitative experimental proposal aims to answer the research questions in section(1.3) that focuses on suggesting new model that mitigate NDP attacks properly by addressing a layered methodology that combines reactive and proactive methodologies minimize the possibility of attacks to occur by using simulation and dataset to do experiments analysis.

The remainder of this chapter is organized as follows: section 3.2 defines the research design. Section 3.3 provides the data sources. Section 3.4 addresses the research instruments . Section 3.5 defines the instrument validation quality . Section 3.6 discusses the data collection procedures . Section 3.7 views the experimental environment implementation 3.8 covers experimental design protocol . Section 3.9 specifies the baseline . Section 3.10 explains the data analysis plan.

3.1 Research path classification(A2-2)

The ML-System(simulation) path is chosen because all measurements are numeric such as detection accuracy rate , CPU overhead%, and latency ms , and they are derived from automated system logs that generated through controlled network simulation(GNS3). There is no direct human interaction involved in the data collection processes. The investigation involves: 1- Always-on Proactive verification (Voucher Based Addressing (VBA) using deterministic key derivation function(KDFs)and shared link Voucher for verifying link layer to IP address binding on every NDP packet) 2- Continuous Reactive ML-based anomaly detection(backpropagation neural network (BPNN) supervised ML algorithm that classifies network traffic as benign or malicious by adjusting internal weights based on error propagation from output layer to input layer and detecting) 3- Layered multi defense architecture(both VBA and BPNN run simultaneously to provide extra protection while maintain low computational cost. Key advantage of VBA(Voucher Based Addressing): unlike literature review models that rely on highly computationally expensive asymmetric or elliptic curve cryptography and PKI infrastructure, VBA employs KDFs and shared link voucher which makes it lightweight, privacy preserving verification mechanism that perfectly suits resource-constrained dual-stack cloud environment. Key advantage of BPNN(backpropagation neural network):it is highly effective for detecting anomalies because it is learning complex, non-linear patterns network traffic. The algorithm uses gradient descent optimization to minimize mean squared error and can generalize from training data to detect novel attacks, even when attacks pattern differ from those have been seen during training.

3.1.1 unit of analysis observation(A2-5)

look to the table below

CONCEPT	DEFINITION
Unit of analysis	Network flow (a sequence of NDP packets (NS/NA/RS/RA)exchange between dual-stack nodes within specific time window of 0.5 ms
Unit of observation	The raw data point collected through a single NDP packet (logged with timestamp/source/destination IP/MAC, and protocol flag)
Run/trial	A complete execution of a defense scenario (1000 NDP packets with 5% spoofing attacks and 3% flooding attacks, executed 15 times per condition to ensure statistical stability.

3.1.2 key metrics (A2-6)

the quantitative experimental research is optimized around the trade-off between detection accuracy and computational overhead. Each metric directly addresses the research question(RQ1:maximize detection and protection rate accuracy while minimizing overhead,RQ2:enable dynamic, real time protection(low latency),RQ3: optimal balance between proactive and reactive components)with special emphasis on CPU efficiency due to VBA'S lightweight design and F1-Score as BPNN's primary optimization metric, and protection rate as the key measure of layered defense effectiveness. The layered model advantage is that the protection rate should be higher than detection rate because VBA blocks attacks even if BPNN misses them.

3.1.3 Early Auditability and Reproducibility Signals(A2-7)

To ensure reproducibility all definitions(metrics) are frozen before implementation, all software versions are locked via docker containers, all ML training uses fixed random seed (42,132) or dataset, all file name follows standardize convention, and all configuration are logged in the evidence pack index

3.1.4 Placement guide (A2-8)

3.2 research design

This study apply a quantitative experimental research by following the network simulation methodology. The design is structured to assist evaluating the defense and protection quality of layered Voucher Based Addressing (VBA) and backpropagation neural network (BPNN) against Neighbor Discovery Protocol (NDP) attacks in dual-stack cloud environment.

3.2.1 Design Translation

1. Design Type: quantitative experimental(network simulation)
2. Unit of Analysis: network flow packets see 2.1.2 unit of analysis
3. Data Source: dataset/logs/simulation output 3.4
4. Groups / Factors/Levels/Scenarios: 4 condition

- Con0: baseline
- Con1:Voucher Based Addressing (VBA) only
- Con2: backpropagation neural network (BPNN) only

• Con3: layered (VBA+BPNN)
5. Key Metrics:
• Protection Rate(PR).
• Detection Rate(DR).
• CPU overhead.
• false positive rate.
• Protection latency.
• F1-Score.
6. Runs/Participants: 15 runs per condition(C0-C3)=60 total run; 0 human participants.
7. Freeze/Lock Decisions:
• docker versions(mininet 2.3.1d , python 3.14)
• Random seeds(42.132)
• Topology script (layeredtopology.py)
• Traffic file(spooftraffic.csv)
• Configuration (experimentalconfig.xlsx)
• Experimental runs(detection-log.csv, performancelog.csv, and audit-trail.csv)
Model training(bpnnv1.pk1)
Crypto implementation(vbav1.py)
BPNN parameters(2 layers,32 neurons, lr=0.01)
VBA parameters(KDF)
Hardware(intel Xeon, 32 GB RAM)
8. Links : section (5,7,8,9,10)

3.3 Design Tables and Mapping Diagrams

3.3.1 RQ-TO-Design Mapping

3.3.2 Groups/Condition table

[a4paper,10pt]article [margin=1in]geometry pdflscape array longtable graphicx booktabs multirow amsmath

3.3.3 GROUP / CONDITION TABLE

metric	Decision rule	target	Operational definition
Protection rate(PR)	Measures overall security effectiveness(RQ1,RQ3):proportion of attacks successfully blocked prevented by the proposed layered model	Target:¿=99.5%	TP BLOCKED/(TP BLOCKED+TP PASSED
Detection rate(DR)	Measures detection efficacy (RQ1,RQ2) proportion of attacks correctly identified by BPNN	Target:¿=99.5%	TP/(TP+FN)
False positive rate(FPR)	Measures false alarm impact (RQ3):proportion of benign traffic incorrectly flagged as malicious	target:¿=99.5%	FP/(FP+TN)
F1-Score	Primary optimization metric (RQ1): harmonic mean of precision and recall (BPNN detecting performance)	Target: ¿=99%	2*(precision *recall)/*(precision +recall)
CPU overhead	Measures computational cost(RQ1,RQ3)CPU utilization (%) during packet processing .VBA advantage lower than SEND and CGA	Minimize CPU overhead	(CPU defense – CPU idle)/ CPU idle *100
Replay Attack [4]	Old NDP messages are resent.	Cache poisoning, DoS, MITM	No specific tool mentioned
Protection latency	Measures response speed (RQ2): time(ms) between malicious packet arrival and successful blocking/mitigation. BPNN classification + VBA verification time included	Target:¡=50ms	Time block taken – time packet receives

signal	How it appears	Decision rule	Early prevention action	evidence
Unit of analysis ambiguity	Vague reference to NDP packet without " without defining the analysis unit (e.g., "we collected 1,000 packets" without specifying flow boundaries)	Must explicitly name Unit of Analysis as Network Flow (sequence of NDP packets exchanged within 1-second window)	Complete A2-5 before any design description	A2-5
Incomplete event definition	inconsistent labeling of "malicious packet" between researchers (e.g., one flags spoofed NS as malicious, another excludes it)	Event definition must be actionable (unify definitions: spoofed NS/NA = malicious, RA flood = malicious, normal NS/NA = benign)	Define events in A2-4 with explicit criteria (packet type + ground truth flag)	A2-4
Metric-RQ misalignment	Generic metric list (e.g., "Detection Rate, CPU, Latency" without linking to specific RQs)	Every metric must be linked to RQ1, RQ2, or RQ3 (e.g., Protection Rate RQ1/RQ3, Protection Latency RQ2)	Complete A2-6 and map RQ Metric (protection rate (RQ1,3) Detection rate (RQ1,2) Protection latency (RQ2, CPU overhead(RQ1,3))	A2-6
Metric change during execution	Modifying metric formula after experiment begins (e.g., changing Detection Rate from TP/(TP+FN) to TP/total)	No modification without documented version control (add Version note to change log with date and rationale)	Create change log with Version notes (e.g., V1.0: Initial metric definitions; V1.1: Added Protection Rate)	Version note section 15)
Unfair comparison	Comparing VBA vs. SEND using different attack intensity (e.g., VBA with 5% spoofing,)	Same source/settings for all conditions (C0–C5): identical Mininet topology, same Scapy traffic replay, same hardware (Intel Xeon, 32GB RAM)	Prepare Fair Comparison Policy (Section 10) with unified scenario and resource budge	Fair comparison policy (SECTION 10)
Non-auditable evidence	Reporting numbers without source (e.g., "Protection Rate = 96%" without linking to log file or run ID)	Every claim must have traceable evidence (each number linked to /logs/detection$_log.csv$+ $runID$+$seedvalue$)	Create Evidence Pack Index (Section 15) listing all config files, dataset splits, and seed values (42, 132) or dataset	Evidence pack index section 15

What is mentioned	placement
introduction	
unit of analysis and observation	(A2-5)
key metrics	(A2-6)
Early Auditability and Reproducibility Signals	(A2-7)

Research Question	Design type	Unit of analysis	Data source	Groups/conditions	Key matrices
RQ1: How can a layered cryptographic verification system ,layering anomaly detection with a lightweight machine learning model ,be optimized to maximize detection while minimizing computational overhead for mitigating NDP spoofing and flooding attacks in resource - constrained dual-stack cloud environment ?	quantitative experimental (network simulation)	Network flow(sequence of NDP packets[NS,NA ,RS,RA])	Dataset/simulation output GNS3 system logs	4 conditions C0- C3	Protection Rate(PR), Detection Rate(DR), CPU overhead, false positive rate.
RQ2: How can the network and Voucher Based Addressing (VBA) be integrated with centralized backpropagation neural network (BPNN) detection engine to enable a dynamic and real-time layered defense against NDP spoofing and flooding attacks in dual-stack cloud environment?	quantitative experimental (time-series analysis)	Network flow(per-packet timestamp measurement for latency calculation	Dataset/simulation output packet logs with timestamp, BPNN predictions	4 conditions C0- C3	Protection latency.
RQ3:what is the optimal balance between proactive cryptographic verification(VBA) and reactive ML- based detection(BPNN) to minimize NDP attacks risk while managing computational overhead in the dual-stack cloud environment ?	quantitative experimental (component isolation)	Network flow(comparative analysis across conditions to identify optimal balance	Dataset/simulation output packet logs with timestamp, BPNN predictions and VBA verification logs+ CPU monitoring	3 conditions (C1:VBA vs C2:BPNN) vs C3:VBA+BPNN)	Protection Rate(PR) trade-off CPU efficiency,F1-Score.

Table 6: Group / Condition Table

Group/Condition	**Description**	**Treatment / Algorithm / Protocol**	**Number of Units (Runs)**	**Notes**
C0	No defense active: standard IPv6 NDP behavior, packets accepted without verification	none	15 runs	Establish attack vulnerability baseline
C1	VBA only: Voucher-Based Addressing verifies link to IP binding on every packet	VBA (KDF)	15 runs	Measure proactive-only blocking effectiveness. Addresses RQ3
C2	BPNN only: backpropagation neural network classifies traffic as benign/malicious	BPNN (2 hidden layers, 32 neurons, lr=0.01)	15 runs	Measure ML-only detection effectiveness. Addresses RQ3
C3	Layered defense: VBA verifies link-to-IP binding + BPNN classifies traffic simultaneously	VBA (KDF) + BPNN	15 runs	Test primary hypothesis (layered advantage). Addresses RQ1 & RQ2

3.3.4 DESIGN TYPE DECISION CARD

Table 7: Design Type Decision Card

Decision	**Selection**	**Justification**
Research type	Quantitative	All measurements are numeric (PR, DR, CPU%, Latency ms)
Methodology	Experimental	Controlled defense condition (C0–C3)
Data source	System logs / dataset	System logs / dataset
Design structure	Pre and post test	Baseline C0 before condition (C1–C3)
Simulation type	Network simulation	GNS3 with topology across conditions
Path classification	A2-2 simulation	Measurement derived from simulation logs

3.3.5 FACTOR LEVELS TABLE

Table 8: Factor & Levels Table

Factor	**Levels**	**Applies to**	**Linked RQs**	**Evidence**	**Note**
F1: Defense condition	A=C0 Baseline; B=C1 VBA only; C=C2 BPNN; D=C3 Layered (VBA+BPNN)	Simulation (GNS3)	RQ1, RQ2, RQ3	Design Translation (3.2.1 item 4), Groups Table (3.2.3)	Primary independent variable (4 levels)
F2: Attack type	A=Spoofing (NS/NA); B=Flooding (RA); C=Benign (normal NDP)	Simulation (Scapy)	RQ1, RQ3	Design Translation (3.2.1 item 7), Spoof_traffic.csv	3 categorical levels
F3: Attack intensity	A=5% spoofing; B=3% flooding	Simulation (Scapy)	RQ1, RQ3	Design Translation (3.2.1 item 7), Confir-v1.xlsx	2 numeric levels
F4: Protection rate (PR)	Continuous (0–100%)	System logs + BPNN predictions	RQ1, RQ3	Design Translation (3.2.1 item 5), Detection_log.csv	Dependent variable: TP_blocked/total
F5: Detection rate (DR)	Continuous (0–100%)	System BPNN predictions	RQ1, RQ2	Design Translation (3.2.1 item 5), Detection_log.csv	Dependent variable: TP/total malicious
F6: CPU overhead	Continuous (0–100%)	System CPU monitoring	RQ1, RQ3	Design Translation (3.2.1 item 5), Performance_log.csv	Dependent variable: Increase over idle
F7: Protection latency	Continuous (ms)	System (packet timestamps)	RQ2	Design Translation (3.2.1 item 5), Audit_trail.csv	Dependent variable: Time_block – Time_receive

3.3.6 DESIGN CONSISTENCY CHECKLIST

Table 9: Design Consistency Checklist

Item	**Rule**	**Verification**	**Evidence**
ICR-1	RQs ↔ Design Type: Each RQ mapped to specific design element	Verified	RQ-Design Mapping Table 3.2.2
ICR-2	All factors have clear, defined levels	Verified	Factor & Levels Table 3.2.5
ICR-3	Groups/Conditions ↔ Metrics: All conditions use same topology, traffic, hardware	Verified	Group/Conditions Table 3.2.3
ICR-4	Runs/Participants ↔ Design Type: Dependent variable match RQ focus	Verified	RQ-Design Mapping Table 3.2.2
ICR-5	Reproducibility: Freeze plan ensure no uncontrolled variation	Verified	Design Freeze & Evidence Plan 3.2.7

3.3.7 DESIGN FREEZE EVIDENCE PLAN

Table 10: Design Freeze & Evidence Plan

Freeze Item	**What Must Be Frozen**	**Why?**	**Decision Rule**	**Evidence**	**Location**	**Ref**
Unit of Analysis	NDP packets (NS, NA, RS, RA)	Reduce ambiguity in PR, DR measurement	No changes after Design Translation	Design Translation + RQ–Design Mapping	3.2.1, 3.2.2	2.2.3
Groups/Conditions	C0: Baseline; C1: VBA only; C2: BPNN only; C3: Layered (VBA+BPNN)	Prevent undefined comparison between VBA vs BPNN vs Layered	No condition without operational definition	Groups / Conditions Table – System	3.2.3	9
Key Metrics	PR, DR, CPU overhead, FPR, Latency, F1-Score	Prevents switching NDP attack metrics after implementation	Fix the metrics before implementation	Design Translation	3.2.1	11
Factor & Levels	F1: 4 levels (C0–C3); F2: 3 types; F3: 2 levels (3%, 5%)	Prevent conflicting NDP attack scenarios	No level without re-executable definition	Factors & Levels Table	3.2.5	9/8
Network topology	2 dual-stack and GNS3	Prevents infrastructure variation in NDP cloud environment	No traffic file changes after freeze-2	Design Translation	3.2.1	–
VBA configuration	KDF, shared link voucher	Prevents inconsistent crypto verification in C1, C3	No VBA changes after freeze-2	Design Translation	3.2.1	–
Fair comparison policy	15 runs per condition, same hardware, same duration	Prevent unfair ML comparison	Policy must be written in Section 10	Fair Comparison Policy + Compute Budget	3.3	10

3.3.8 DESIGN-LINKED VALIDITY SIGNALS LOG

Table 11: Design-Linked Validity Signals Log

Validity Threat	**Design Countermeasure**	**Evidence**	**Reference**
Bias: different conditions use different topology	Identical topology (6 dual-stack nodes across all conditions)	Factors & Levels Table	3.2.5
External factors affect the result	Controlled simulation environment	Design Translation	3.2.1
Testing effect	Pre-test/post-test design with independent baseline	Groups / Conditions Table	3.2.3

3.3.9 PLACEMENT GUIDE

Table 12: Placement Guide

Guide Section	**Chapter 3 Section**	**Summary**	**Reference**
Design Type	First item within section 3.2.1	Design type name and operational reason	11
Unit of Analysis	First item within section 3.2.1	Operational definition and unit type	2.2.3
Design Translation	Independent paragraph in section 3.2.1	Filled template	8,9,10,11,13,15
RQ–Design Mapping Table	Table within section 3.2.2	RQ coverage (3 RQ mapped)	4,11
Groups/Conditions	Table within section 3.2.3	Condition definition (C0–C3)	9,19
Factors & Levels	Table 3.2.5	All factors	11
Freeze & Evidence Plan	Table within section 3.2.7	What will be frozen and how documented	15,16

3.4 DATA SOURCE DECISION CARD

Table 13: Data Source Decision Card

Item	**Specification**	**Justification**
Source path	ML-System (Simulation)	All measurements are numeric and derived from automated system logs
Primary data source	GNS3 network simulation output	Controlled environment, reproducible, no human subject
Secondary data source	Synthetic attack traffic (Scapy-generated)	Allow controlled attack injection for spoofing and flooding scenario
Data format	CSV (Detection_log, Performance_log, Audit_trail)	Standard format for quantitative analysis
Ground truth	Labeled attack traffic (spoofed NS/NA, flooded RA)	Enables supervised learning for BPNN
Unit of analysis	Network flow (sequence of NDP packets within 1-second window)	Defined in section 3.1.2
No human participation	0 participants	Fully automated simulation

3.4.1 DATASET DESCRIPTION

Table 14: Dataset Description

Traffic Type	**Description**	**Samples per Run**	**Label**
Normal NDP	Legitimate NS, NA, RS, RA messages	2000	Benign (0)
Spoofing attack	Forged NS/NA messages (MITM)	500	Malicious (1)
Flooding attack	Excessive RA messages (DoS)	300	Malicious (1)
Total per run	–	**2800**	–
Total (15 runs × 4 conditions)	–	**168,000**	–

3.4.2 DATASET COMPOSITION (SPLIT)

Table 15: Dataset Composition (Split)

Dataset Split	**Percentage**	**Samples**	**Purpose**
Training set	70%	117,600	BPNN training
Validation set	15%	25,200	Hyperparameter tuning
Test set	15%	25,200	Final evaluation

3.4.3 FEATURES OF BPNN

Table 16: Features of BPNN

Feature	**Description**	**Type**
Message type	NS, NA, RS, RA	Categorical
Source IP Address	Source of NDP message	Numeric
Destination IP Address	Destination of NDP message	Numeric
Packet length	Size of NDP packet (byte)	Numeric
Inter-arrival time	Time since last packet (ms)	Numeric
Hop limit	IPv6 hop limit value	Numeric
ICMPv6 code	ICMPv6 code value	Numeric
Source MAC Consistency	Whether MAC matches expected	Binary

3.4.4 DATA SOURCE SELECTION SCORECARD

Table 17: Data Source Selection Scorecard

Criterion	**Why It Matters**	**Decision Rule**
Task alignment	Ensures measurement matches RQs	If task/label match RQ → accept
Unit of analysis	Prevent aggregation errors	If unit unclear → no final selection
Label Quality	Directly affects validity	If label undocumented → reject
Class balance	Prevent misleading metrics	If imbalance severe → document policy
Recency/representation	Prevent reality gap	If outdated → note as threat
License/access	Prevent illegal use	If no clear license → do not use
Privacy	Ethical/legal compliance	If PII present → protection plan
Leakage check	Prevent inflated performance	If future features exist → remove
Reproducibility	Enables replication	If version cannot be retrieved → do not use
Compute budget	Ensures feasibility	If exceeds budget → justify sample

3.4.5 RQ-TO-SOURCE MAPPING MATRIX

Table 18: RQ-to-Source Mapping Matrix

RQ	**Variable**	**Data Source**	**Unit of Analysis**	**Evidence**	**Decision Rule**
RQ1	PR, DR, CPU overhead	Detection_log.csv, Performance_log.csv	Network flow	TP_block/total attacks	If unit unclear → no final selection
RQ2	DR, Protection latency	Detection_log.csv, Audit_trail.csv	Network flow	TP/total malicious, Time_block − Time_receive	If latency not measured → rerun
RQ3	CPU overhead, PR, DR	Detection_log.csv, Performance_log.csv	Network flow	CPU increase over baseline, TP_block/total attacks	If imbalance → document policy

3.5 INSTRUMENT INVENTORY

Table 19: Instrument Inventory

Instrument	**Purpose**	**Type**	**Version**	**Linked RQs**	**Output**
GNS3	Network simulation	Simulation	2.2.51	RQ1, RQ2, RQ3	Network topology, packet flow
Scapy	Packet crafting and attack generation	Traffic generation	2.4.6	RQ1, RQ3	Spoofed NS, NA, flooded RA
THC-IPv6 Toolkit	NDP attack tool (flood_router, flood_solicitate)	Attack tool	3.6	RQ1, RQ3	Flooding attack traffic
Python	Scripting and data processing	Programming	3.14	All RQs	Logs, metrics, BPNN
TensorFlow/Keras	BPNN implementation	ML framework	2.2	RQ1, RQ2	Detection prediction
Wireshark/tcpdump	Packet capturing and analysis	Monitoring	4.6.9	RQ1, RQ2, RQ3	Packet traces
Docker	Containerization for reproducibility	Environment	4.95.0	All RQs	Version-locked environment
psutil	CPU/memory monitoring	Performance	Latest	RQ1, RQ3	CPU overhead, memory usage

3.5.1 METRIC DEFINITION CARDS

Table 20: Metric Definition Cards

Element / Metric	**PR**	**DR**	**FPR**	**F1-Score**	**CPU Overhead**	**Protection Latency**
Definition	Proportion of attacks successfully blocked/prevented by the proposed layered model	Proportion of attacks correctly identified by BPNN	Proportion of benign traffic incorrectly flagged as malicious	Harmonic mean of precision and recall	Percentage increase in CPU utilization compared to baseline (C0)	Time taken to block/verify an NDP packet
Computation	PR = TP_BLOCKED / (TP_BLOCKED + TP_PASSED)	DR = TP / (TP + FN)	FPR = FP / (FP + TN)	F1 = 2×(P×R)/(P+R)	CPU_overhead = (CPU_cond − CPU_base)/CPU_base × 100	Latency = Time_block − Time_receive (ms)
Time window	Per experimental run (5-min attack window)	Per experimental run	Per experimental run	Per experimental run	Per experimental run	Per packet
Unit of analysis	Network flow	Network flow	Network flow	Network flow	System-level measurement	Network flow
Data source	Detection_log.csv	Detection_log.csv	Detection_log.csv	Detection_log.csv	Performance_log.csv	Audit_trail.csv
Target	≥ 99.5%	≥ 99.5%	≤ 0.5%	≥ 99%	≤ 15%	≤ 10 ms
Limitation	May be affected by attack intensity	Depends on BPNN training quality	Trade-off with detection rate	Primary optimization metric for BPNN	Hardware-dependent	Affected by system load
Linked RQ	RQ1, RQ3	RQ1, RQ2	RQ3	RQ1	RQ1, RQ3	RQ2
Implementation notes	TP_BLOCKED = attacks blocked by VBA or BPNN; TP_PASSED = attacks that bypassed defense	TP = malicious correctly classified; FN = malicious missed	FP = benign flagged malicious; TN = benign correctly classified	Precision = TP/(TP+FP); Recall = TP/(TP+FN)	CPU_baseline = avg CPU during C0; CPU_cond = avg CPU during C1–C3	Time_receive = packet arrival; Time_block = block timestamp

3.5.2 INSTRUMENT-VARIABLE MAPPING

Table 21: Instrument–Variable Mapping

Variable / Concept	**Instrument**	**Number of Items**	**Data Type**	**Scoring Method**	**Note**
PR	VBA + BPNN + Detection_log.csv	1 metric	Ratio %	TP_BLOCKED / total attacks	Layered defense effectiveness
DR	BPNN + Detection_log.csv	1 metric	Ratio %	TP / (TP + FN)	BPNN-only detection
FPR	BPNN + Detection_log.csv	1 metric	Ratio %	FP / (FP + TN)	False alarm impact
F1-Score	BPNN + Detection_log.csv	1 metric	Ratio %	2×(P×R)/(P+R)	Primary optimization metric
CPU overhead	System monitor (psutil) + Performance_log.csv	1 metric	Ratio %	(CPU_cond − CPU_base)/CPU_base × 100	Computational cost
Protection latency	Packet timestamps + Audit_trail.csv	1 metric	Continuous (ms)	Time_block − Time_receive	Real-time responsiveness

3.6 INSTRUMENT DEVELOPMENT STRATEGY

Table 22: Instrument Development Strategy

Strategy	**Definition**	**Application in This Study**
Adopted	Use existing tool as-is	GNS3, Wireshark, Docker, Python used as-is with documented versions
Adapted	Modify existing tool for context	Scapy attack scripts – adapted from public NDP attack examples for dual-stack cloud
Developed	Build new tool from scratch	BPNN model, VBA module, Decision Fusion Engine – developed specifically for this study

3.6.1 INSTRUMENT DEVELOPMENT STRATEGY (DETAILED)

Table 23: Instrument Development Strategy (Detailed)

Instrument	**Strategy**	**Justification**	**Change Made**
GNS3	Adopted	Standard network simulation platform	None, used as-is
Scapy	Adapted	Public NDP attack scripts exist	Adapted for dual-stack cloud topology, added spoofing/flooding parameters
THC-IPv6	Adopted	Standard NDP attack toolkit	None, used as-is
BPNN model	Developed	No existing model for NDP anomaly detection in dual-stack cloud	Built with TensorFlow/Keras, 2 layers, 32 neurons, lr=0.01
VBA Module	Developed	VBA not previously implemented in cloud simulation	Built with Python, KDF-based voucher generation
Decision Fusion Engine	Developed	No existing fusion logic for VBA + BPNN	Built with Python, weighted voting (VBA=0.6, BPNN=0.4)
Logging System	Developed	Custom event schema for NDP defense	Built with Python, 6 event types

3.6.2 PILOT STUDY PLAN

Table 24: Pilot Study Plan

Element	**Specification**
Purpose	Validate simulation setup, attack generation, and measurement system
Sample	3 runs per condition (C0–C3) = 12 pilot runs
Duration	1 week
Data collected	Detection_log, Performance_log, Audit_log
Success criteria	All logs generated correctly; attack traffic matches expected patterns; BPNN training converges; no system crashes; metrics within target ranges
Decision rules	If logs incomplete → fix logging; if attacks fail → adjust Scapy scripts; if BPNN doesn't converge → tune hyperparameters; if metrics outside range → investigate and adjust

3.6.3 VALIDITY TYPES

Table 25: Validity Types for ML-System Research

Validity Type	**Adapted for ML-System**	**How to Verify in This Study**
Face validity	Tools appear to measure what they claim	Expert review, topology verification
Content validity	Features cover all attack dimensions	Feature coverage matrix (Table 3.5.5)
Construct validity	Model learns meaningful patterns	Training/validation loss, confusion matrix
Criterion validity	Model meets target thresholds	Compare PR, DR, FPR against targets
Ecological validity	Simulation represents real NDP traffic	Realistic attack intensities

3.6.4 FEATURE COVERAGE MATRIX

Table 26: Feature Coverage Matrix

Attack Type	**Relevant Features**	**Coverage**
Spoofing (NS/NA)	Message type, Source IP, Destination IP, Source MAC consistency	Covered
Flooding (RA)	Message type, Inter-arrival time, Packet length, Hop limit	Covered
Benign NDP	All 8 features (baseline pattern)	Covered

3.6.5 VALIDITY EVIDENCE MAP

Table 27: Validity Evidence Map

Concept / Variable	**Validity Type**	**How Collected**	**When**	**Expected Output**
PR	Construct + Criterion	Comparison with baseline (C0), target threshold	Pilot + Main	Significant improvement over baseline, $\geq 99.5\%$
DR	Construct + Criterion	Comparison with BPNN-only (C2), target threshold	Pilot + Main	Layered $>$ individual, $\geq 99.5\%$
FPR	Criterion	Measurement against target	Pilot + Main	$\leq 0.5\%$
F1-Score	Construct + Criterion	Standard ML metric, target threshold	Pilot + Main	$\geq 99\%$
CPU overhead	Criterion	Measurement against baseline	Pilot + Main	$\leq 15\%$
Protection latency	Ecological + Criterion	Realistic NDP traffic, target threshold	Pilot + Main	≤ 10 ms

3.6.6 RELIABILITY PLAN

Table 28: Reliability Plan

Reliability Type	**Adapted for ML-System**	**How Verified**
Internal consistency	Consistent metric calculation across conditions	Same formula applied to C0–C3
Test-retest	Same experiment run multiple times	15 runs per condition with fixed seeds (42, 132) for BPNN training
Inter-rater reliability	Automated logging consistency	100% consistent (no human rater)
Stability	Consistent results across runs	Compare mean ± SD across 15 runs

3.7 DATA COLLECTION FLOW MATRIX

Table 29: Data Collection Flow Matrix

Phase	**Objective**	**Source / Tool**	**Unit**	**Time**	**Output**	**Control Notes**
T0	Baseline measurement	GNS3 + Wireshark	Network flow	Before attacks	Baseline_traffic.csv	No attacks, normal NDP
T1	Attack injection	Scapy / THC-IPv6	Network flow	During experiment	Attack_traffic.csv	Controlled intensity (5% spoofing, 3% flooding)
T2	Defense evaluation	VBA + BPNN	Network flow	During attacks	Detection_log.csv	Same topology across C0–C3
T3	Performance measurement	System monitor (psutil)	System	During experiment	Performance_log.csv	Same hardware
T4	Latency measurement	Packet timestamps	Network flow	During attacks	Audit_trail.csv	Timestamps synchronized
T5	Data export & verification	Python script	–	After each run	Combined_dataset.csv	Verify completeness, log incidents

3.7.1 COLLECTION PROTOCOL (STEPS)

Table 30: Collection Protocol

Step	**Action**	**Details**
5*Step 1: Pre-run setup	Verify GNS3 topology loaded	6 dual-stack nodes, all reachable
	Confirm Docker containers running	Python 3.14, TensorFlow, Scapy, THC-IPv6
	Check logging enabled	Detection_log, Performance_log, Audit_trail
	Initialize random seeds	42, 132 for BPNN training
	Verify instrumentation readiness	All 6 events logging correctly
5*Step 2: Condition execution	Load condition configuration	C0: Baseline, C1: VBA only, C2: BPNN only, C3: Layered
	Start packet capture	Wireshark/tcpdump on monitoring port
	Launch attack traffic	Scapy (spoofing NS/NA), THC-IPv6 (flooding RA)
	Monitor defense system	VBA verification + BPNN prediction
	Record metrics	PR, DR, FPR, F1-Score, CPU, Latency
4*Step 3: Data export & verification	Stop packet capture	After 5-minute attack window
	Export logs	Detection_log.csv, Performance_log.csv, Audit_trail.csv
	Verify completeness	Check all events logged, no missing PID/SID
	Record deviation	Log any incident in Incident Log

3.7.2 QUALITY CONTROL

Table 31: Quality Control

Check	**Frequency**	**Success Criteria**	**Action if Failed**
Log completeness	Per run	100% of expected events logged	Re-run, fix logging
Attack intensity	Per run	Within ±10% of configured	Adjust Scapy/THC-IPv6
BPNN convergence	Per run	Training loss decreases	Tune hyperparameters
Metric stability	Across runs	SD $< 5\%$	Investigate variance
System stability	Per run	No crashes	Restart, document incident

3.7.3 INCIDENT HANDLING

Table 32: Incident Handling

Incident	**Decision Rule**	**Action**
Logging fails	If critical event missing	Re-run; document in Incident Log
Attack intensity wrong	If deviation $> 10\%$	Adjust parameters; re-run
System crash	If GNS3/Docker crashes	Restart; re-run; document
BPNN doesn't converge	If loss increases after 50 epochs	Tune hyperparameters; re-run
Metric outside target	If PR, DR, FPR deviate	Investigate; adjust threshold

3.8 ENVIRONMENT TOPOLOGY

Table 33: Environment Topology

Elements	**Selection**	**Justification**
Environment type	Simulation (GNS3)	Controlled, reproducible, no production risk
Isolation	VMware (GNS3) environment	Prevent interference with production
Reproducibility	Docker container	Ensure consistent environment across runs
Network topology	Dual-stack cloud (6 nodes)	Represents small-to-medium cloud environment

3.8.1 HARDWARE BASELINE

Table 34: Hardware Baseline

Component	**Specification**	**Role**
Host machine	Intel Xeon, 32 GB RAM	Runs GNS3 and all VMs
Storage	500 GB SSD	Stores simulation data and logs
Network	1 Gbps Ethernet	Internal simulation network

3.8.2 IMPLEMENTATION SNAPSHOT

Table 35: Implementation Snapshot

Elements	**Specification**
System / platform	GNS3 network simulation
Purpose	Evaluation of layered NDP defense (VBA + BPNN)
Build / version	GNS3, Docker image tagged
Attack scenarios	Spoofing (NS/NA), Flooding (RA)
Configuration	4 conditions (C0–C3)
Feature flags	VBA ON/OFF, BPNN ON/OFF
Seed / test data	70/15/15 split, fixed seeds (42, 132)
Fault handling	Restart run, document in Incident Log
Key components	VBA module, BPNN module, Decision Fusion Engine

3.8.3 FREEZE DECISION PLAN

Table 36: Freeze Decision Plan

Item	**What Must Be Frozen**	**Decision Rules**
Random seed	42, 132	No changes after freeze
Topology script	layered_topology.py	No modifications during collection
Traffic file	spoof_traffic.csv, flood_traffic.csv	No changes to attack patterns
BPNN parameter	2 layers, 32 neurons, lr=0.01	No tuning during evaluation
VBA parameters	KDF configuration	No changes to KDF settings
Hardware	Intel Xeon, 32 GB RAM	Same hardware for all runs

3.8.4 PRE-RUN VALIDATION

Table 37: Pre-Run Validation

Item	**Verification**	**Status**
Build / tag review	Check Docker image tag	☐
Config setting	Compare critical setting	☐
Feature flags	Check ON/OFF list	☐
Data loading	Load seed/test data	☐
Login test	Test two roles	☐
Event logging	Verify 6 events log correctly	☐
Timezone / time sync	Verify timestamps	☐
Connection / latency	Measure baseline latency	☐
Failure points review	Identify potential failures	☐

3.9 DATA ANALYSIS PLAN

Table 38: Data Analysis Plan

Focus	**Key Rules / Action**	**Output**
Design to analysis mapping	Map RQ1 → ANOVA (C0–C3), RQ2 → ANOVA (C1–C3), RQ3 → Regression	Analysis Mapping Matrix
Data preparation & cleaning	Remove duplicates (PID + Timestamp + Event); unify units (sec, MB); UTC timezone; link PID + SID + RunID	Cleaned dataset
Missing & outlier policy	Primary missing → exclude; <20% missing → impute; Outliers → IQR flag, do not delete	Missing/Outlier policy doc
Derived metrics	PR = TP_BLOCKED/total; DR = TP/(TP+FN); FPR = FP/(FP+TN); F1 = 2PR/(P+R); CPU = (CPU_cond − CPU_base)/CPU_base × 100; Latency = Time_block − Time_receive	Variable Dictionary
Descriptive statistics	Mean, Median, SD, IQR for all metrics; Boxplot, Bar chart, Histogram, Line chart	Descriptive summary tables
Assumption checks	Normality (Shapiro-Wilk) → Non-parametric; Homogeneity (Levene) → Welch; Sphericity → Greenhouse-Geisser	Assumption-to-Decision tables
Statistical tests	Between → One-way ANOVA + Tukey; Ordinal → Kruskal-Wallis + Dunn; Within → RM-ANOVA; Binary → Chi-square; Regression → Linear	Test results
Effect size & CI	Cohen's d, dz, Odds Ratio, Partial η^2, Cliff's delta; report 95% CI for primary outcomes	Effect size table
Multiple comparisons	Limit 1–2 primary outcomes per RQ; Bonferroni/Holm/FDR for pre-defined families	Multiplicity control plan
Sensitivity analysis	Re-run with/without outliers; non-parametric	Sensitivity plan table